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Evidence for significant clockwise rotations of the Korean Peninsula during Cretaceous

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ABSTRACT

In an effort to evaluate the Cretaceous magnetostratigraphy for the Korean Peninsula and to establish the tectonic coherence of its various elements, we collected paleomagnetic data from 121 samples from 20 sites within the Chilgok Formation (108.3–109.9 Ma) in the Gyeongsang Basin. Together with previously published data, we evaluate the results from a total of 163 sites in the basin.

We combine our age model with results from recent stratigraphic, paleomagnetic and radiometric geochemical studies. In this study, we found that two distinct declination shifts decrease with younging direction, indicating two clockwise rotational events of the Korean Peninsula with respect to the Eurasia continent. The earlier event took place during 130–100 Ma (Phase I, newly termed "Goguryeo Disturbance") and a later one during 80–50 Ma (Phase III, belonging to "Bulguksa Orogeny"). The mean rotation rate in the interval from 115.2 to 103.8 Ma (Phase I) is about 0.74°/Ma, while the rate from 90.9 to 79.8 Ma (Phase II) is 0.19°/Ma. Based on paleolatitude change during Phase I, we infer that the Korean Peninsula (eastern part of the Sino-Korea Block) migrated southward about 300 km after the complete amalgamation of the Sino-Korea Block into the Eurasian continent resulting in N–S compression within the Korean peninsula and Manchuria. Large-scale strike-slip faults (e.g., Tan-Lu Fault, Okcheon Boundary Fault) were probably rejuvenated in the Sino-Korea Block during Phase I.

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1. Introduction

The Gyeongsang Basin is located in the southeastern part of the Korean Peninsula and is composed of the Milyang, Uiseong, and Yeongyang sub-basins (Fig. 1). Previous paleomagnetic analysis of the Yeongyang sub-basin (Park et al., 2005) led to the suggestion that this sub-basin underwent a counterclockwise rotation by 16.3° ± 4.6° with respect to the other two sub-basins. Park et al. (2005) proposed that the rotation of the Yeongyang sub-basin was caused by the sinistral movement of the basin-bounding fault during the latest Cretaceous to Paleogene.

The previous studies lumped both the magnetite-carried and hematite-carried directions together and lacked any detailed field tests to distinguish between primary and secondary magnetizations. In addition, the full scale magnetostratigraphy has not been reported in the Cretaceous sequence, which is essential to evaluate the tectonic evolution of the Korean Peninsula (Kim et al., 1993a; Zhao et al., 1999; Park et al., 2005). In order to address these problems, a paleomagnetic analysis of the Milyang and Uiseong sub-basins is critical so as to obtain a reliable and representative Cretaceous paleopole for the Korean Peninsula. Furthermore, a detailed paleomagnetic study of the formations in the Gyeongsang Supergroup is essential to constrain the direction and timing of magnetization of the Gyeongsang Basin.

The magnetostratigraphy of the Gyeongsang basin is poorly constrained in part due to poor age constraints. Recently, reliable U–Pb radiometric age data from zircon grains in the sedimentary and volcanic rocks were obtained (e.g. Jwa et al., 2009; Lee et al., 2010b). Using this new age control and the sequential stratigraphic interpretation of the Gyeongsang Basin (Chough and Sohn, 2010), we perform a reconnaissance paleomagnetic study of the Chilgok Formation, which unconformably overlies the Precambrian crystalline basement.

2. Geologic setting of the Gyeongsang Basin

The Cretaceous Gyeongsang Supergroup, the largest supracrustal belt in the Korean Peninsula, is composed of three groups: the Shindong (characterized by little or very small amount of volcanic rock fragments), the Hayang (containing volcanic rocks and rock fragments), and the Yucheon (mainly consisting of intrusive and extrusive rocks) Groups in stratigraphically ascending order. The sedimentary sequences generally dip eastward with dip angles between 0° and 30° (Fig. 1).

The Shindong Group is considered to have formed in an extensional backarc trough (Chough and Sohn, 2010) and consists of the Nakdong Formation (conglomerate intercalated with sandstone and mudstone), Hasandong Formation (combination of red to pale green sandstone and shale), and Jinju Formation (gray to dark gray

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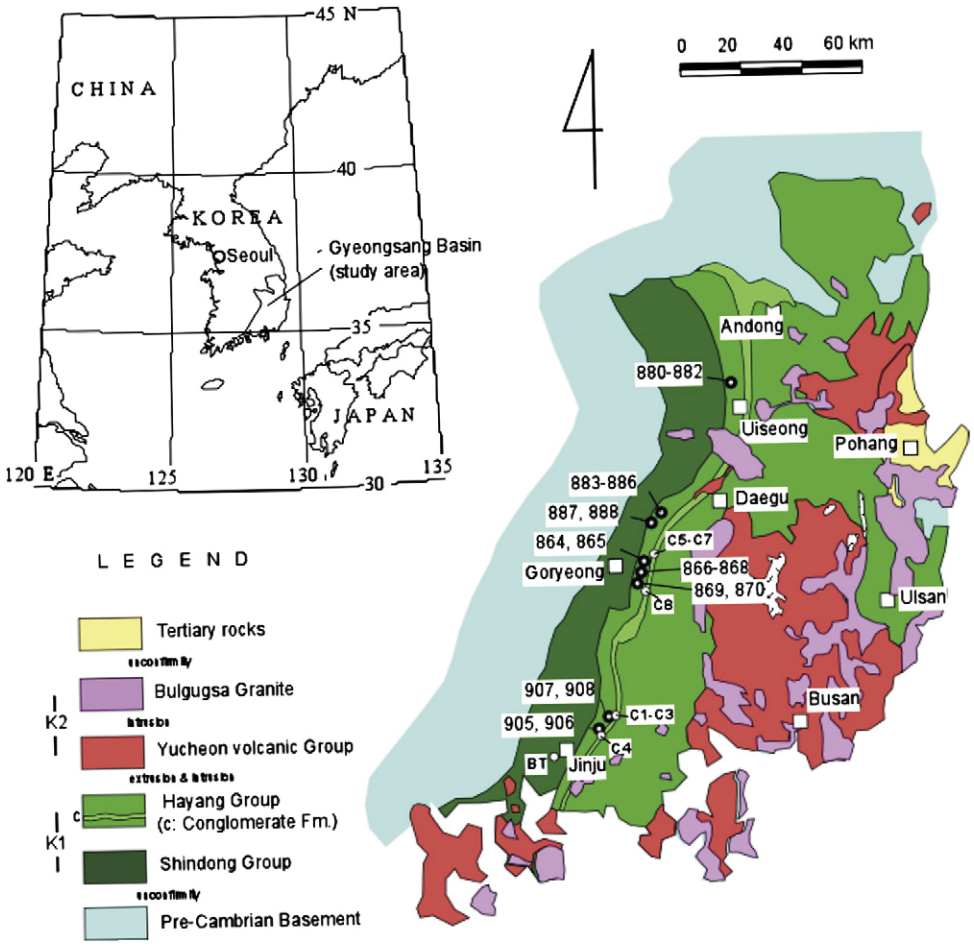


Fig. 1. Geologic map with paleomagnetic sampling sites (Sites 864 to 908). Labels of 'C1 to C8' and 'BT' denote sites of conglomerate test (Kim et al., 1998) and baked test (Jeon et al., 1998).

shale and sandstone). The total thickness of the Shindong Group is about 2000–4000 m (Lee, 1987).

The Hayang Group (Fig. 1) varies in thickness from about 2000 to 5000 m (Lee, 1987), and consists of sandstone, shale, conglomerate, marl and tuff (Lee, 1987). The timing of deposition of this group is debated due partly to a poor fossil record.

The red beds of the Gyeongsang Supergroup contain abundant volcanic rock fragments. The Chilgok Formation, the lowest part of the Hayang Group, is red in color and contains a variety of mafic to felsic volcanic rock fragments (>33%) as well as granite and quartzite fragments (Lee, 1987).

3. Reconnaissance paleomagnetic study of the Chilgok Formation

3.1. Sampling and laboratory procedures

The Chilgok Formation is intercalated in the middle part of the Gyeongsang Supergroup, and is well recognized in the field. Samples for the pilot study were mostly collected from the red beds. Rocks in the vicinity of intrusive dikes or in hydrothermal alteration zones were avoided because they might have been affected by heating and thermochemical transformations. Five to nine samples were collected at each sampling site, covering 1–3 m stratigraphic thickness. The core samples were later cut in the laboratory to obtain standard specimens of 24 mm diameter, and then sliced to 23 mm length to prepare for measurements. A total of 121 paleomagnetic samples were collected from 20 sites.

Natural remanent magnetization (NRM) was measured using a cryogenic SQUID rock magnetometer and a spinner rock magnetometer. All specimens were demagnetized by stepwise heating in a non-inductive electric furnace enclosed in four nested shields. The residual field in the furnace during cooling cycle was about 5 nT. Low-field susceptibility was measured using a Bartington MS2 susceptibility meter after each step of the progressive thermal demagnetization in order to detect chemical changes due to heating. The progressive thermal demagnetization data were analyzed using the principal component analysis (PCA) method (Kirschvink, 1980), and site means were calculated using Fisher's statistics (Fisher, 1953). Thermomagnetic analysis and isothermal remanent magnetization (IRM) acquisition were applied to representative samples to study magnetic minerals that carry the remanent magnetization. Thermomagnetic analysis was performed by using an automatic recording semi-horizontal thermobalance. Petrographic inspection of polished surfaces and thin-sections of selected specimens was also carried out under reflected and transmitted light.

3.2. Results

One of the main objectives of this study is to reveal the characteristic magnetization in the red beds of the Chilgok Formation. Typical thermomagnetic results in this study suggest magnetite and hematite to be the predominant magnetic minerals in the samples. The unblocking temperatures of about 570°–575 °C and 675°–680 °C confirm the presence of magnetite and hematite, respectively. The irreversible portion of the cooling cycle may result from the creation of new ferromagnetic minerals from the paramagnetic and superparamagnetic materials due to recrystallization during heating (Fig. A1).

The initial natural remanent magnetization (NRM) intensity for red bed specimens is about 10^{-2} A/m. The initial NRM direction is well grouped towards N to NNE and downward. All specimens were then demagnetized by stepwise heating, and their remanent directions were analyzed using the PCA.

3.2.1. Component A

This component is positively inclined in situ with generally north to northwest directions, or is not sufficiently defined to allow calculation of a meaningful direction. This component contributes little to the total NRM of the red bed. Its unblocking temperature does

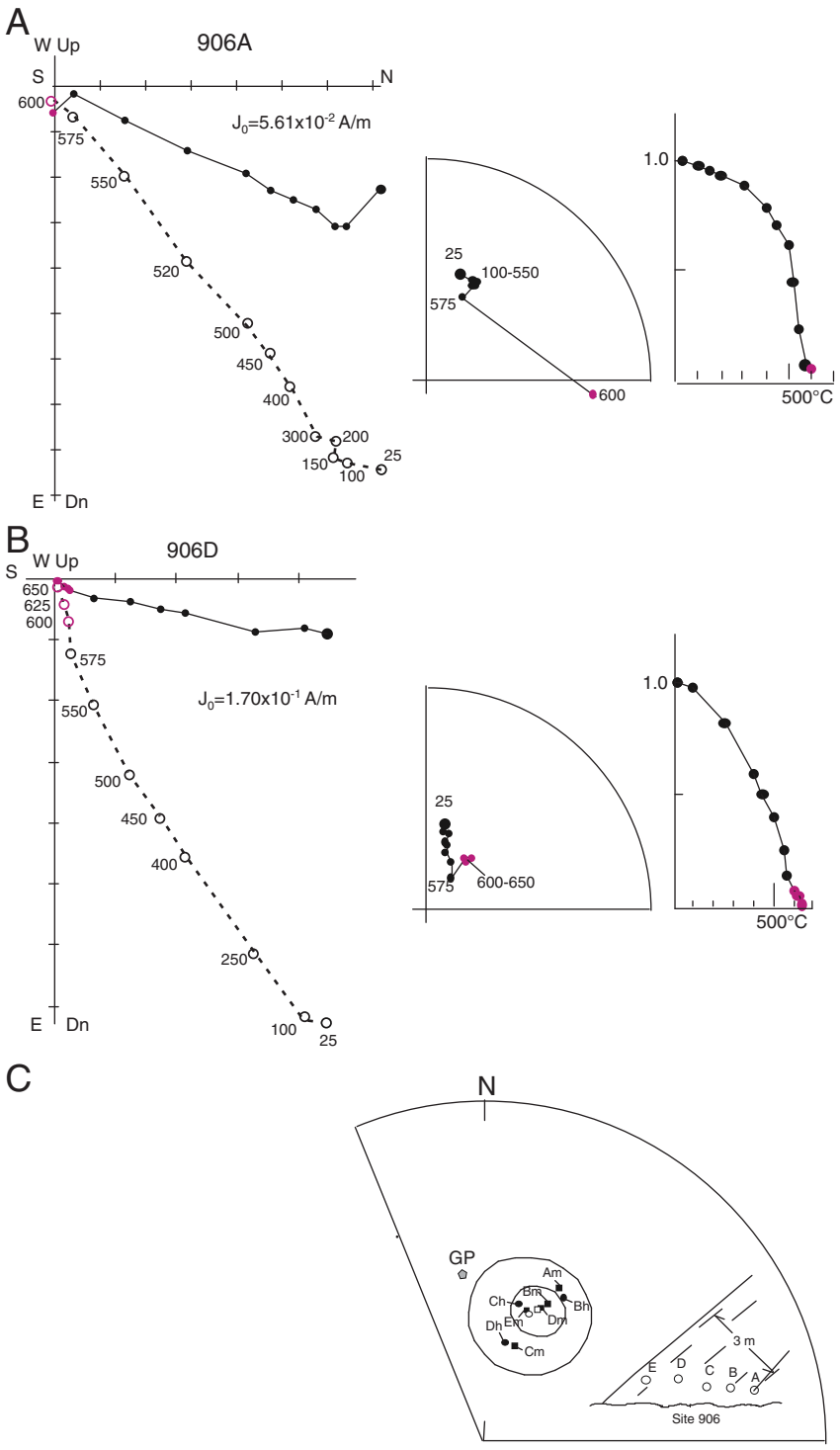


Fig. 2. (A and B) Zijderveld plots with demagnetization and equal area projections of stepwised thermal demagnetized specimens (*in-situ*), showing single component (906A, left) of magnetite and dual components (906D, right) of magnetite and hematite. Solid and open circles are the projection onto horizontal and vertical planes respectively. Jo is the initial NRM (Natural Remanent Magnetization) intensity value. The red symbols in color show the behavior of the higher unblocking temperature than 585 °C. (C) Equal area diagram (in situ) of magnetite (square) and hematite (circle) characteristic components of Site 906. Solid square and circle are the site mean directions of magnetite and hematite component respectively. All symbol plots on the lower hemisphere. GP denotes present geomagnetic direction in the site. The letter with subscript represents the specimen name with magnetic component. Note that the site mean direction of both magnetite and hematite component are agreed together at 95% confidence level.

not exceed 500 °C. We interpret its origin as a viscous remanence of present Earth magnetic field or laboratory contamination.

3.2.2. Components B and C

The components of magnetization can be distinguished by their unblocking temperature spectra and directions. These form the dominant magnetization of the red bed residing in magnetite (Component B) and hematite (Component C) grains.

Typical demagnetization results show directions determined for the characteristic components of magnetite and hematite in five specimens from Site 906 (Fig. 2). The univectorial magnetizations from specimens A and E were revealed with fairly discrete unblocking temperatures in the range of 550° to 575 °C, indicating magnetite as the principal magnetic remanence carrier. No higher unblocking temperature magnetization can be determined due to erratic demagnetization behavior after 575 °C (Fig. 2A). Additional specimens B, C, and D from this site reveal multi-component characteristic magnetizations of magnetite (Component B, 250°–575 °C) and hematite (Component C >575 °C). As shown in Fig. 2B (specimen 906D), two linear segments are identified by PCA method after removal of viscous remanent magnetization (VRM) at less than 250 °C. The characteristic direction of Component B within a specimen is slightly different from that of Component C, however, the site mean direction of Component B (open square in Fig. 2C) for 5 specimens (A, B, C, D, and E) is indistinguishable from that of Component C (open circle in Fig. 2C) for 3 specimens (B, C, and D) at 95% confidence level (Fig. 2C).

Each sample was treated by stepwise thermal demagnetization, and the result was classified into Component B (magnetite, <575 °C) and Component C (hematite, >575 °C) by the unblocking temperature spectra as shown in Fig. 3. The results from this pilot study show mean declination and inclination of 30.4° and 57.0° ($\alpha_{95}=2.1°$) for Component B and 30.3° and 55.9° ($\alpha_{95}=2.1°$) for Component C, and 30.5° and 56.5° ($\alpha_{95}=1.9°$) when Components B and C are combined. The various results (Fig. 3) are statistically indistinguishable after bedding correction (Table 1).

The fold test for the mean directions of both components is also illustrated by the equal area plots before and after bedding correction (Fig. 3). The improved clustering of the mean direction for both Components B and C is evident from the increase in the precision parameter for the tilt-corrected mean (Fig. 3).

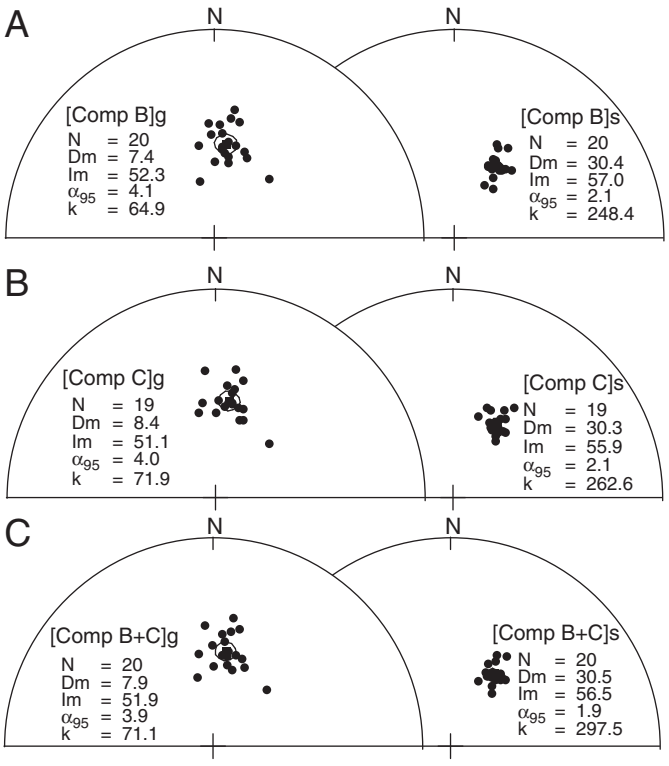


Fig. 3. Equal area projection of site mean direction with 95% cone of confidence in geographic (left) and stratigraphic (right) coordinates. (A) Component B of magnetite, (B) component C of hematite, and (C) Components B and C. N denotes number of sites. All projections are on the lower hemisphere.

3.3. Origins of dual magnetic Components B and C in the Gyeongsang Basin

In this pilot study, both Components B and C show a very high concentration between sites ($\alpha_{95}=1.9°$) and pass the fold test ($k_2/k_1=4.18$; $n=20$) at 95% (1.72) and 99% (2.16) confidence levels (McElhinny, 1964). Stepwise unfolding for both Components B and C shows optimum clustering at 90–100% unfolding, indicating that both are pre-fold magnetization in origin (Fig. 4). The observed polarity of both components are all normal, indicating that these magnetizations were acquired during the Cretaceous Long Normal Superchron (KLNS) ranging from 124.6 to 84.0 Ma (Walker and Geissman, 2009). Though bedding attitudes are not significantly different among sites, the directions of Components B and C for the study area are well clustered in the northeasterly and intermediate downward direction after bedding correction.

Zhao et al. (1999) reported that the same characteristic direction is carried by both magnetite and hematite, suggesting that hematite probably formed by auto-oxidation processes during the primary cooling of the basalts in the Gyeongsang Basin. This is evidently supported by the conglomerate test for volcanic cobbles from the Shilla Conglomerate Formation (Kim et al., 1998), showing that the ChRMs of both magnetite and hematite with parallel to subparallel directions both passing the fold test. Photomicrographic observations show that the hematite is a replacement product of the pre-existing mineral in the same basalt cobble.

Kim (1988) also reported characteristic magnetizations of hematite and magnetite, and interpreted that both magnetization are DRM or post-DRM as a syn-depositional origin. Petrographic observations suggest that a large number of detrital grains in the Chilgok Formation came from volcanic rocks (Chang, 1977; Choi, 1986; Chough and Sohn, 2010). MPMS (magnetic property measuring system) results for the red bed samples display the existence of Verway transition of magnetite at 120 K (Zhao et al., 1999). We concluded that Component B of magnetite and the Component C of hematite are primarily acquired during DRM or post-DRM origin from the Gyeongsang volcanic arc system (Chough and Sohn, 2010).

3.4. Reliability evaluation on the pilot study

The paleomagnetic results from the Chilgok Formation are evaluated by the following seven reliability criteria (Van der Voo, 1990):

(1) A well determined formation age and the contemporaneous magnetization: The Chilgok Formation is easily recognizable in the field with good lateral continuity along the basin margin. The stable characteristic remanence of the formation may not have been remagnetized by thermochemical process as attested to the positive tests at 95% confidence level. We interpret that the timing of magnetization in the pilot sites range from 108.3 ± 1.1 Ma to 109.9 ± 0.5 Ma (Fig. 5).

(2) Sufficient number of samples and adequate statistical precession (yes): The characteristic direction (Components B and C) was determined from 20 sites (121 samples), and the parameters k and α_{95} are 297.5 and 1.9°, respectively.

Table 1
Paleomagnetic result for the pilot study of the Chilgok Formation in the Gyeongsang Basin, Korea (see Fig. 2).

Site	N_b	Component A							Component B							Components A + B						
		(n)	D_g	I_g	D_s	I_s	α_{95}	k	(n)	D_g	I_g	D_s	I_s	α_{95}	k	(n)	D_g	I_g	D_s	I_s	α_{95}	k
Goryeong(128.4°E 35.7°N)																						
864	7	(5)	8.3	51.0	27.2	58.0	9.6	64.4	(5)	9.7	52.3	29.7	58.8	5.0	232.2	(10)	9.0	51.7	28.5	58.5	4.6	112.0
865	7	(5)	42.1	58.0	32.8	57.1	7.6	101.7	(5)	44.4	59.5	34.4	58.8	8.4	84.8	(10)	43.2	58.8	33.5	58.0	4.8	102.4
866	7	(6)	8.1	41.5	25.9	50.0	8.4	64.7	(4)	9.9	46.0	30.9	53.5	8.9	107.9	(10)	8.8	43.3	27.8	51.4	5.5	79.5
867	8	(5)	356.6	43.7	27.7	55.2	8.0	91.4	(4)	354.7	38.5	20.5	51.9	11.9	60.3	(9)	355.7	41.4	24.4	53.8	5.9	77.5
868	6	(5)	0.2	59.7	26.3	55.1	8.1	89.5	(5)	359.7	56.3	23.0	52.3	5.3	208.5	(10)	359.9	58.0	24.6	53.7	4.2	131.6
869	9	(5)	344.5	66.8	29.6	66.1	6.1	157.9	(4)	348.4	55.5	17.0	56.3	8.0	131.7	(9)	346.6	61.8	23.0	61.9	5.5	87.1
870	7	(4)	357.8	48.7	24.7	49.0	8.1	128.3	(2)	2.1	52.3	30.2	49.3	38.3	44.6	(6)	359.2	49.5	26.6	49.1	6.6	103.3
Daegu (128.4°E 35.8°N)																						
880	8	(5)	9.0	37.3	32.0	56.4	12.0	41.9	(5)	9.2	37.6	32.5	56.6	6.9	124.9	(10)	9.1	37.5	32.3	56.6	5.8	70.6
881	8	(4)	2.7	44.5	28.2	58.6	9.7	91.0	(6)	6.2	44.9	32.5	57.6	7.4	83.1	(10)	4.8	44.8	30.9	58.0	5.0	94.3
882	7	(5)	5.4	54.7	38.2	65.1	9.4	67.4	(6)	7.5	51.9	36.7	62.1	6.9	94.0	(11)	6.6	53.2	37.4	63.5	5.0	85.8
883	7	(4)	13.3	52.2	35.1	56.7	8.5	118.2	(5)	14.5	53.0	37.0	57.0	6.7	130.3	(9)	14.0	52.7	36.2	56.9	4.3	141.7
884	7	(6)	4.4	47.7	30.5	48.3	5.6	143.7	(4)	8.9	47.5	34.1	46.5	5.0	336.3	(10)	6.2	47.7	32.0	47.7	3.5	191.3
885	7	(5)	349.9	52.7	25.0	58.2	6.7	130.9	(3)	351.7	51.9	25.6	56.9	9.1	183.4	(8)	350.6	52.4	25.2	57.7	4.3	166.7
886	7	(4)	19.4	53.7	32.9	57.8	8.5	117.6	(5)	17.5	52.9	30.4	57.3	13.2	34.6	(9)	18.4	53.3	31.6	57.6	6.9	56.4
887	7	(5)	11.9	42.7	33.0	61.6	8.8	77.2	(6)	13.2	41.4	34.1	60.5	6.5	106.5	(11)	12.6	41.8	33.7	61.1	4.6	100.6
888	7	(3)	5.0	53.3	26.1	58.4	12.9	92.6	(3)	6.1	53.7	27.6	58.5	10.7	134.1	(6)	5.5	53.5	26.8	58.5	5.8	136.4
Jinju (128.2°E 35.2°N)																						
905	9	(6)	10.3	59.8	25.4	58.7	3.4	395.3	(7)	17.0	57.9	30.5	55.9	3.6	282.7	(13)	14.0	58.9	28.3	57.3	2.4	289.4
906	5	(5)	22.4	55.7	39.8	54.8	6.3	148.4	(3)	19.6	57.4	38.4	57.0	14.5	72.9	(8)	21.4	56.3	39.3	55.6	5.0	125.3
907	8	(5)	9.7	57.3	35.8	55.8	4.3	322.2	–	–		–		–		(5)	9.7	57.3	35.8	55.8	4.3	322.2
908	9	(5)	7.2	56.1	34.9	57.1	7.5	103.9	(5)	9.7	51.7	33.2	52.5	8.4	84.9	(10)	8.6	53.9	34.1	54.8	5.0	95.8
$Mean_g$		(20)	7.4	52.3			4.1	64.9	(19)	8.4	51.1			4.0	71.9	(20)	7.9	51.9			3.9	71.1
$Mean_s$		(20)			30.4	57.0	2.1	248.4	(19)			30.3	55.9	2.1	262.6	(20)			30.5	56.5	1.9	297.5

N_b (n), number of individual block samples (components); D_g (D_s), site mean declination in geographic (stratigraphic) coordinate; I_g (I_s), site mean inclination in geographic (stratigraphic) coordinate; $Mean_g$ ($Mean_s$), result mean direction in geographic (stratigraphic) coordinate.

(3) Adequate demagnetization (yes): All data was determined by vector subtraction given appropriate thermal interval (Zijderveld, 1967; Kirschvink, 1980).
(4) Field test (yes): The characteristic direction passes a fold test. A baked contact test (Jeon et al., 1998) and a conglomerate test (Kim et al., 1998) in the vicinity of the sampling sites (Fig. 1) indicate no indication of remagnetization.
(5) Structural control and tectonic coherence (yes): The basin formed on the Precambrian metamorphic basement in tensional to trans-tensional stress field back-arc system with virtual absence of deformation, syntectonic folding or faulting (Chough and Sohn, 2010).
(6) The presence of reversals (no): In this study no reversals were found, suggesting that the formation may have developed during a short duration ranging from 108.3 ± 1.1 Ma to 109.9 ± 0.5 Ma based on the age model. Therefore this interval belongs to the CLNS (Cretaceous Long Normal Superchron).
(7) Resemblance to paleopoles of younger age (no): The direction with high declination and inclination values (30.5° and 56.5°) has not been reported from the Late Cretaceous and Cenozoic formations in Korea.

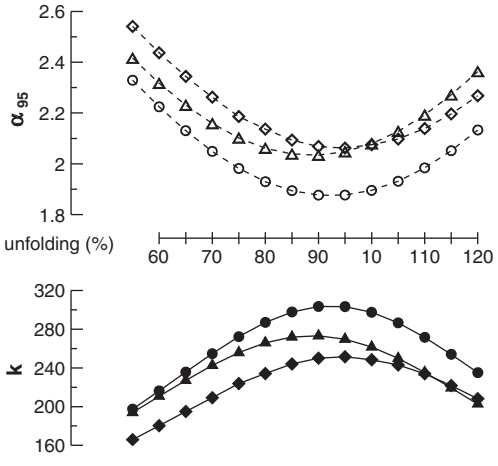


Fig. 4. Unfolding test for precision parameter k (solid symbol with solid line) and α_{95} (open symbol with dotted line) during progressive unfolding. The diamonds, triangles, and circles represent Components B (magnetite), C (hematite), and B + C (both magnetite and hematite), respectively.

4. Compilation of Cretaceous paleomagnetic data in Korea

A characteristic remanence was reported from the main part of the Gyeongsang Basin (Milyang and Uiseong sub-basins). In these sub-basins, we selected 163 data sites from 15 results including this study by applying the following criteria (Table 2): (1) each dataset includes at least 4 samples; (2) a 95% confidence interval less than 15°; (3) successful thermal and/or alternating field demagnetization; (4) absence of remagnetization; (5) available sites for the age of magnetization (for example, existence of age data for igneous rocks); and (6) data from the stable Milyang and Uiseong sub-basins showing distinct stratigraphic position with gentle dips toward east to southeast. Despite the monoclinal structure of the formations, all results of selected sites except some of the granite sites (with no referenced dip value) pass the fold test at the 95% confidence level.

4.1. Age model for the Shindong and Hayang Groups

All of the selected paleomagnetic sites were revisited in order to investigate the relative position of the sedimentary units in stratigraphic succession (i.e., lower, middle, or upper), or, in case of the igneous rocks, their isotopic ages. The lithostratigraphic classification of the Gyeongsang Supergroup is well established by Chough and Sohn (2010). In this study, we improved their time-stratigraphic succession of the basin using an age model based on

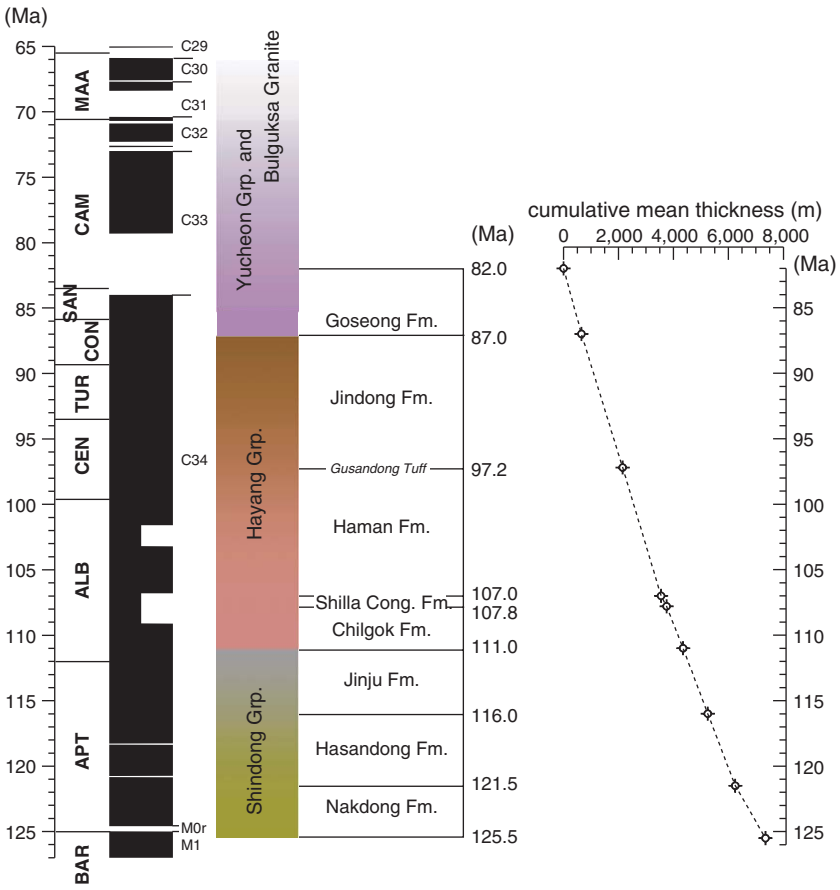


Fig. 5. Stratigraphy (left) and cumulative mean deposition rate (right) of the main formations in the Gyeongsang Supergroup in Korea. See text for further explanations. Abbreviations: BAR, Barremian; APT, Aptian; ALB, Albian; CEN, Cenomanian; TUR, Turonian; CON, Coniacian; SAN, Santonian; CAM, Campanian; MAA, Maastrichtian.

a combination of radiometric data and magnetostratigraphy (Fig. 5).

In our age model, a thin pyroclastic density current deposit (1–5 m in thickness) named the Gusandong tuff is adopted as the most important key bed because of (1) very thin and continuous outcrop throughout the basin, (2) reliable U–Pb zircon age data (97.1 ± 2.0 Ma and 97.3 ± 1.8 Ma; Jwa et al., 2009), and (3) very short extrusion period. We assign the eruption age as 97.2 Ma.

Lee et al. (2010b) reported U–Pb zircon SHRIMP ages from the lowest part of the Jinju and Shilla-Conglomerate formations. The youngest radiometric ages obtained from the Jinju and Shilla-Conglomerate formations are 112.4 ± 1.3 Ma and 110.4 ± 2.0 Ma, respectively, indicating the lower limit of their formation ages. This age (110.4 ± 2.0 Ma) is supported by ^{40}Ar–^{39}Ar age (113.4 ± 2.4 Ma, hornblende from a basalt cobble; Kim et al., 2005) for the Shilla Conglomerate formation. We note that the ages concentrate around 110 to 113 Ma, implying that these volcanic materials might have been derived from a volcanic arc system (Chough and Sohn, 2010).

The stratigraphic sequences of the Shindong and Hayang Group and the lower part of Yucheon Group exhibit a dominantly normal polarity, which is recognized in KLNS (Cretaceous Normal Superchron, C34n). Some reversed polarity signatures are sporadically found in the Haman and Jeomgok (Lee et al., 1987; Kim et al., 1993b; Zhao et al., 1999), Hasandong (Kim et al., 1993a), and Nakdong (Kim et al., 1993a) Formations. These facilitate the magnetostratigraphy in this study area. For example, dual polarities with successful reversal test were found in the Haman formation (Zhao et al., 1999). Combining with the U–Pb age of the overlying Gusandong tuff, the depositional age of the Haman Formation with dual polarity may be assigned to the Albian including brief mixed subchrons of $C34_{1r}$ (118.26 Ma) and $C34_{2r}$ (120.74 Ma). On the other hand, both normal and reversed polarities after KLNS are generally recognized in the Yucheon Group and Bulguksa Granites, which agree well with their radiometric ages (Walker and Geissman, 2009).

4.2. Stratigraphic age of the sedimentary sequence

For the age estimates from the sedimentary rocks, we provide the relative stratigraphic mean position from the base (m/n, m ≤ n) for each sampling site with respect to each formation (Table 2). This method may be applicable only for the monocline formations like that of Gyeongsang Basin.

The relative sedimentary age (T) and the error (dT) may be estimated using the equation.

$$T = T_L - (T_L - T_U) * (2\,m - 1) / 2n, dT = (T_L - T_U) / n$$

where T_L and T_U denote the lower-limit formation age and the upper-limit formation age, respectively.

Assuming that two sampling sites of middle (Site A) and upper (Site B) parts of a Formation, m/n values are 2/3 and 3/3, respectively. Assuming T_L and T_U for the Formation are given 109 Ma and 100 Ma, then, sedimentation ages for Sites A and B are 104.5 ± 3.0 Ma and 101.5 ± 3.0 Ma, respectively.

4.3. Radiometric age for the Yucheon Group and Bulguksa granite

The Late Cretaceous igneous rocks which are exposed in the eastern part of the Gyeongsang Basin, are seen to extrude over the Hayang Group or intrude into the Shindong or Hayang Group. Shin

Table 2
Cretaceous paleomagnetic data from the Gyeongsang Basin, Korea.

Site		Site sampled			Geographic		Stereographic		Parameters			VGP position					Ref	Model or radiometric			Formation Name	Rock type
List	Name	Lat(N)	Long(E)	n_b	D_g	I_g	D_s	I_s	α_{95}	k	Δd	Lat(N)	Long(E)	dp	dm	(N/R)	[no.]	Age(Ma)	±	Method (m/n)		
001	B5	35.8	129.3	4	191.4	−55.8	–	–	4.3	480.0	7.7	80.8	212.1	4.4	6.1	R	[3]	55.3	2.9	K–Ar, bt[a]	Bulguksa Granite	Granite
002	AW	36.2	128.9	6	18.6	55.2	–	–	6.7	101.1	11.8	75.0	215.5	6.8	9.5	N	[6]	65.8	1.5	K–Ar, K-fd[b]	Bulguksa Granite	Granite
003	DJ	35.7	129.2	4	192.2	−51.5	–	–	8.5	120.0	13.7	79.3	235.1	7.9	11.6	R	[3]	66.3	4.0	K -Ar, bt[a]	Bulguksa Granite	Granite
004	OA	35.6	129.0	5	357.8	52.7	–	–	2.7	830.0	4.5	87.0	347.7	2.5	3.7	N	[3]	66.3	4.0	K -Ar, bt[a, c]	Bulguksa Granite	Granite
005	OB	35.6	129.0	6	18.3	62.1	–	–	8.3	67.0	18.0	73.9	184.5	10.1	13.0	N	[3]	66.3	4.0	K -Ar, bt[a]	Bulguksa Granite	Granite
006	OT	35.6	129.0	6	3.0	50.4	–	–	5.1	175.0	8.0	84.9	278.9	4.6	6.8	N	[3]	66.3	4.0	K -Ar, bt[a]	Bulguksa Granite	Granite
007	OU	35.6	129.1	4	16.6	50.4	–	–	4.8	390.0	7.5	75.4	231.9	4.4	6.5	N	[3]	66.3	4.0	K -Ar, bt[a]	Bulguksa Granite	Granite
008	C27	35.8	129.1	4	3.6	54.6	–	–	8.7	116.0	15.1	87.0	229.8	8.7	12.3	N	[3]	66.3	4.0	K–Ar, bt[a]	Bulguksa Granite	Granite
009	Site 4	35.9	129.0	6	6.6	66.3	–	–	5.1	142.8	12.8	76.3	147.3	6.9	8.4	N	[12]	68.1	3.4	K–Ar, wr[d]	Hakbong Vol. Member	Basalt
010	AK	36.3	128.7	5	189.5	−54.2	–	–	12.3	18.4	21.4	82.1	227.0	12.1	17.3	R+N	[6]	69.1	3.3	K–Ar, wr[b]	Yucheon Group	Acidic tuff
011	AC	36.1	128.8	7	192.1	−54.7	–	–	8.7	49.0	15.2	80.1	220.2	8.7	12.3	R	[6]	69.1	3.3	K–Ar, wr[b]	Yucheon Group	Acidic tuff
012	GPS	34.9	128.3	4	214.2	−44.4	205.5	−56.0	7.2	86.9	13.0	73.3	206.7	7.4	10.3	R	[2]	69.8	2.4	$C31^2$	Yucheon Group	Welded tuff
013	DP	35.5	129.2	4	5.6	46.4	–	–	11.7	65.0	17.1	80.8	276.5	9.7	15.1	N	[3]	72.0	1.4	K–Ar, bt[a, c]	Bulguksa Granite	Granodiorite
014	DQ	35.5	129.2	4	31.9	52.5	–	–	14.1	46.0	23.6	63.7	215.2	13.4	19.5	N	[3]	72.0	1.4	K–Ar, bt[a, c]	Bulguksa Granite	Granodiorite
015	GG	34.9	128.3	8	43.7	46.5	–	–	3.3	280.3	4.8	52.3	216.8	2.7	4.2	N	[2]	72.8	3.5	$C32^2$	Yucheon Group	Andesite
016	GK	34.9	128.3	7	27.7	55.9	–	–	4.5	184.2	8.0	67.5	206.3	4.6	6.5	N	[2]	72.8	3.5	$C32^2$	Yucheon Group	Andesite
017	GA	35.0	128.3	5	208.6	−51.3	211.8	−42.3	9.4	35.8	12.8	60.6	230.6	7.1	11.6	R	[2]	72.8	3.5	$C32^2$	Yucheon Group	Andesite
018	GB	35.0	128.3	4	193.7	−4.5	208.5	−39.0	10.1	84.2	13.0	61.9	238.2	7.2	12.0	R	[2]	72.8	3.5	$C32^2$	Yucheon Group	Andesite
019	DR	35.5	129.1	4	36.5	52.3	–	–	14.3	44.0	23.8	59.9	213.2	13.5	19.6	N	[3]	75.0	–	K–Ar, bt[a]	Bulguksa Granite	hb-bt granite
020	DT	35.5	129.1	4	9.3	53.1	–	–	13.0	53.0	22.0	82.1	229.4	12.5	18.0	N	[3]	75.0	–	K–Ar, bt[a]	Bulguksa Granite	hb-bt granite
021	JS1	35.7	128.7	7	333.4	38.0	346.2	50.3	5.2	135.0	8.2	77.6	20.7	4.7	7.0	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
022	JSS1	35.7	128.7	6	356.2	51.1	18.6	55.8	5.4	157.0	9.6	75.0	210.8	5.5	7.7	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Tuffaceous ss.
023	JS4	35.7	128.7	8	9.9	49.3	27.7	55.2	5.5	103.0	9.7	67.6	210.4	5.6	7.8	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
024	JS3	35.7	128.7	4	346.5	43.9	17.2	56.1	6.3	213.0	11.3	76.1	209.7	6.5	9.1	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
025	JS6	35.7	128.7	6	356.2	49.9	15.3	52.3	6.3	113.0	10.3	77.1	226.8	5.9	8.6	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
026	JS2	35.7	128.7	4	12.0	45.5	30.0	46.2	8.8	110.0	12.8	63.3	228.0	7.2	11.3	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
027	JSS2	35.7	128.7	4	344.2	47.6	3.2	56.1	9.8	36.0	17.8	87.2	197.5	10.1	14.1	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Tuffaceous
028	JS5	35.7	128.7	6	350.4	48.5	8.4	52.4	9.8	47.0	16.2	82.6	237.6	9.3	13.5	N	[10]	76.7	4.3	$C33.n^{10, a, e}$	Jusasan Fm.	Andesite
029	JS11	35.8	129.1	12	176.3	−54.6	202.1	−65.1	4.9	81.0	11.7	71.2	175.8	6.4	7.9	R	[10]	81.2	4.7	C33	Jusasan Fm.	Andesite
030	GI'	35.7	129.3	5	187.0	−39.9	211.2	−58.4	11.1	46.6	21.6	65.1	202.0	12.2	16.4	R+N	[1]	83.6	11.4	Rb–Sr, wr[f]	Jusasan Fm.	Tuffaceous ss/sh
031	GJ	34.9	128.3	9	24.9	60.9	9.4	54.2	4.3	145.1	7.4	82.3	216.9	4.2	6.0	N	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuff
032	GQ	35.0	128.3	9	50.5	53.1	36.0	55.5	7.6	46.7	13.5	60.8	205.7	7.7	10.9	N	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuffaceous ss
033	GH	34.9	128.3	11	36.4	53.9	29.3	52.4	7.9	34.2	13.0	65.7	214.4	7.5	10.9	N	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuffaceous ss/sh
034	GI	34.9	128.3	10	22.1	48.8	23.1	52.4	8.6	32.8	14.2	70.8	217.4	8.1	11.8	N+R	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuffaceous ss
035	GO	34.9	128.3	5	216.2	−50.5	211.6	−59.6	9.5	65.0	19.0	64.5	196.3	10.7	14.3	R	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuffaceous sh
036	GN	34.9	128.3	4	184.8	−56.3	203.3	−52.6	10.0	85.2	16.6	70.7	216.7	9.5	13.8	R	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuffaceous ss/sh
037	GF	34.9	128.3	8	21.6	57.8	44.8	63.1	2.6	456.2	5.8	54.7	188.6	3.2	4.1	N	[2]	84.5	5.0	(1/1)	Goseong Fm.	Tuff
038	CDDh	35.1	129.0	13	33.2	65.1	34.2	38.3	9.6	19.7	12.3	57.2	234.2	6.7	11.4	N+R	[4]	84.5	1.0	(1/5)	Dadaepo Fm.	ss/sh
039	CDDm	35.1	129.0	15	9.8	67.9	31.8	40.2	6.6	34.2	8.7	59.8	234.2	4.8	8.0	N	[4]	84.5	1.0	(1/5)	Dadaepo Fm.	ss/sh
040	CDD	35.1	129.0	10	2.6	70.8	30.0	46.6	6.7	52.4	9.8	63.5	226.4	5.5	8.6	N	[4]	84.5	1.0	(1/5)	Dadaepo Fm.	ss/sh
041	CR'	35.9	128.9	5	25.1	53.5	18.4	61.5	4.9	248.9	10.3	74.3	188.0	5.8	7.6	N	[1]	87.8	1.7	(3/3)	Geoncheonri Fm.	Gray ss
042	CS'	36.0	128.6	4	22.0	50.6	23.1	57.5	3.6	640.2	6.7	71.5	205.2	3.9	5.3	N	[1]	87.8	1.7	(3/3)	Geoncheonri Fm.	Gray ss
043	DN	35.7	129.2	4	18.5	59.2	–	–	8.8	115.0	17.4	74.8	197.3	9.9	13.2	N	[3]	89.0	10.0	K–Ar, wr[a]	Bulkuksa Granite	Granodiorite
044	MY22	35.4	128.3	7	358.9	59.3	17.9	45.5	5.8	109.8	8.3	72.6	242.1	4.7	7.4	N	[8]	90.8	2.6	(3/4)	Jindong Fm.	ss/sh
045	GF'	35.7	129.4	6	22.7	46.5	19.6	60.1	6.3	115.8	12.7	73.8	194.5	7.2	9.5	N	[1]	90.9	0.7	(2/7)	Geoncheonri Fm.	Gray ss
046	JIND	35.8	128.6	(14)	15.4	48.4	28.6	53.5	4.9	71.5	8.3	66.6	214.5	4.8	6.8	N	[14]	92.1	10.2	(1/1)	Jindong Fm.	Sedimentary rock
047	MY21	35.4	128.3	8	354.5	59.7	16.5	48.7	9.4	36.0	14.3	75.0	236.0	8.1	12.4	N	[8]	92.1	2.0	(3/5)	Jindong Fm.	ss/sh
048	MY20	35.4	128.3	8	350.2	59.0	14.4	51.7	4.1	180.4	6.6	77.7	228.4	3.8	5.6	N	[8]	93.0	1.7	(3/6)	Jindong Fm.	ss/sh
049	16	35.8	129.0	5	39.7	48.2	28.4	51.3	6.5	145.0	10.4	66.2	219.9	6.0	8.8	N	[11]	93.8	0.5	(3/4)	Songnaedong Fm.	Red sh/ms
050	18	36.1	127.9	12	2.7	43.5	15.2	51.7	6.6	44.5	10.7	76.9	230.1	6.1	9.0	N	[11]	94.2	0.3	(3/6)	Songnaedong Fm.	Red sh/ms
051	GC'	36.1	128.5	7	29.4	58.3	29.5	55.8	8.6	50.7	15.4	66.3	209.2	8.8	12.3	N	[1]	94.6	0.3	(2/7)	Songnaedong Fm.	Gray ss
052	46	36.3	128.9	7	31.9	52.0	20.9	51.3	8.0	58.0	12.9	72.2	226.9	7.4	10.9	N	[7]	94.6	1.0	(3/5)	Chunsan Fm.	ss/sh
053	45	36.3	128.9	7	22.0	40.7	34.4	57.1	6.2	94.5	11.5	62.6	205.5	6.6	9.0	N	[9]	94.6	1.7	(2/3)	Chunsan Fm.	ss/sh
054	CW	35.1	129.2	7	58.9	51.3	25.0	49.1	4.0	176.9	6.1	68.3	227.3	4.0	6.0	N	[3]	95.3	1.6	(8/9)	Daegu Fm.	Red ss

055	CV	35.7	129.2	10	71.5	50.1	33.4	55.4	6.4	59.9	11.3	63.0	208.7	6.4	9.0	N	[3]	95.3	1.6	(8/9)	Daegu Fm.	Red ss
056	CX	35.7	129.2	7	76.0	66.6	25.8	57.2	7.2	80.9	13.4	69.1	204.2	7.2	9.8	N	[3]	95.3	1.6	(8/9)	Daegu Fm.	Red sh
057	AE	36.2	128.8	11	72.4	61.1	30.7	47.3	4.4	107.1	6.5	63.0	226.8	3.7	5.7	N	[6]	95.9	2.6	(1/2)	Chunsan Fm.	Red sh
058	44	36.3	128.9	10	8.5	48.4	11.5	54.5	3.2	235.5	5.5	80.6	223.1	3.2	4.5	N	[9]	95.9	2.6	(1/2)	Chunsan Fm.	ss/sh
059	17	36.0	128.5	11	7.9	49.5	17.3	52.2	5.6	68.0	9.2	75.4	225.9	5.3	7.7	N	[11]	96.1	0.7	(2/3)	Banyaweol Fm.	Red sh/ms
060	OS	35.7	129.2	7	58.9	54.8	28.8	51.5	6.2	81.4	10.0	65.9	219.3	6.2	9.1	N	[3]	96.9	1.6	(7/9)	Daegu Fm.	Reddish green sh
061	OQ	35.7	129.2	8	56.6	64.7	53.4	53.4	6.8	62.2	11.5	71.7	218.1	6.8	9.8	N	[3]	96.9	1.6	(7/9)	Daegu Fm.	Reddish green sh
062	2	36.3	128.9	6	18.6	40.9	37.3	64.6	7.8	74.4	18.4	60.5	187.0	10.1	12.5	N	[9]	97.2	2.1	U–Pb, zr[g]	Gusandong Fm.	Tuff
063	46	36.3	128.9	12	6.0	38.0	10.6	57.0	7.3	36.0	13.5	81.4	206.9	7.7	10.6	N	[9]	97.2	2.1	U–Pb, zr[g]	Gusandong Fm.	Tuff
064	YF	35.5	129.1	5	36.0	37.9	14.1	50.9	4.9	151.6	7.8	77.6	233.7	4.9	7.3	N	[3]	98.4	1.6	(6/9)	Daegu Fm.	Greenish gray ss
065	YA	35.5	129.1	5	30.5	51.8	13.4	59.8	5.9	144.7	11.8	76.2	188.9	5.9	7.9	N	[3]	98.4	1.6	(6/9)	Daegu Fm.	Greenish red ss
066	YE	35.5	129.2	7	30.8	51.7	19.3	60.1	5.0	145.6	10.1	73.9	193.2	5.0	6.6	N	[3]	100.0	1.6	(5/9)	Daegu Fm.	Green ss/red sh
067	YB	35.5	129.2	7	23.7	45.2	15.7	55.3	7.4	53.2	13.1	77.3	213.0	7.4	10.3	N	[3]	100.0	1.6	(5/9)	Daegu Fm.	Greenish gray ss
068	YD	35.5	129.2	7	39.4	43.9	29.4	51.3	3.0	276.4	4.8	65.3	219.0	3.0	4.5	N	[3]	101.6	1.6	(4/9)	Daegu Fm.	Green ss/red sh
069	YC	35.5	129.2	7	32.8	50.0	20.1	55.5	4.7	130.3	8.3	73.7	211.3	4.7	6.6	N	[3]	101.6	1.6	(4/9)	Daegu Fm.	Greenish red ss
070	MY17	35.5	128.3	8	29.4	73.2	36.2	57.5	6.6	71.5	12.4	61.1	202.1	7.1	9.7	N	[8]	102.1	9.8	(1/1)	Haman Fm.	ss/sh
071	15	35.9	128.6	6	27.2	56.6	47.4	57.5	8.4	66.0	15.8	52.5	200.6	9.0	12.3	N	[11]	102.1	3.3	(2/3)	Haman Fm	Red sh
072	Ham	35.8	128.4	(7)	11.2	66.0	27.6	59.5	3.9	285.3	7.7	67.9	198.2	4.4	5.9	N	[14]	102.1	9.8	(1/1)	Haman Fm	Sedimentary rock
073	Ham A	35.9	128.6	20	8.3	56.2	356.4	62.7	3.1	114.0	6.8	81.4	111.1	3.8	4.9	N	[15]	102.1	9.8	(1/1)	Haman Fm	Red ss/slts
074	Ham B	35.9	128.6	23	18.6	44.3	27.7	51.3	5.2	31.1	8.3	66.8	220.3	4.8	7.1	N+R	[15]	102.1	9.8	(1/1)	Haman Fm	Red ss/slts
075	14−	36.2	128.3	4	214.0	−35.2	221.0	−38.1	5.5	290.0	7.0	51.5	229.4	3.8	6.5	R	[11]	104.6	4.9	(1/2)	Haman Fm.	Red sh
076	14+	35.9	128.5	5	10.4	38.5	17.2	45.0	8.3	88.0	11.8	72.6	246.3	6.6	10.5	N	[11]	104.6	4.9	(1/2)	Haman Fm.	Red sh
077	MY18	35.5	128.3	7	12.9	45.1	17.2	57.6	7.1	74.0	13.3	76.0	201.8	7.6	10.4	N	[8]	104.6	1.6	(2/6)	Haman Fm.	ss/sh
078	MY27	35.3	128.3	7	14.1	58.1	21.7	53.4	7.6	64.4	12.8	72.1	216.3	7.4	10.6	N	[8]	104.6	4.9	(1/2)	Haman Fm.	ss/sh
079	22	36.2	127.8	10	34.6	49.6	44.5	54.2	5.0	115.8	8.6	54.0	207.6	4.9	7.0	N	[9]	104.6	1.6	(2/6)	Sagok Fm.	ss/sh
080	43	36.3	128.8	9	17.5	47.6	20.7	59.6	4.0	62.7	7.9	73.3	198.2	4.5	6.0	N	[9]	104.9	1.4	(2/7)	Sagok Fm.	ss/sh
081	49	36.3	128.8	9	24.2	43.4	35.4	59.3	2.6	897.0	5.1	62.1	200.1	2.9	3.9	N	[9]	104.9	1.4	(2/7)	Sagok Fm.	ss/sh
082	HA	35.1	127.9	10	16.6	52.2	33.2	55.4	3.5	189.3	6.2	63.1	206.4	3.6	5.0	N	[5]	105.4	3.3	(1/3)	Haman Fm.	ss/sh
083	MY26	35.3	128.3	7	17.3	64.8	35.5	61.0	7.2	71.7	15.0	61.7	193.7	8.4	11.0	N	[8]	105.4	3.3	(1/3)	Haman Fm.	ss/sh
084	MY25	35.3	128.3	8	22.2	62.1	25.6	49.3	9.7	33.7	15.0	67.9	224.5	8.5	12.9	N	[8]	105.8	2.5	(1/4)	Haman Fm.	ss/sh
085	MY24	35.3	128.3	11	7.0	70.1	40.4	58.8	3.1	223.1	6.0	57.9	198.5	3.4	4.6	N	[8]	106.0	2.0	(1/5)	Haman Fm.	ss/sh
086	AG	36.2	128.8	7	80.2	69.3	32.7	58.6	7.9	60.1	15.3	64.1	201.9	8.7	11.7	N	[6]	106.3	1.4	(1/7)	Sagok Fm.	Red sh
087	MY23	35.5	128.3	5	350.4	67.6	22.7	58.9	7.6	102.6	14.8	71.5	198.0	8.4	11.3	N	[8]	106.3	1.4	(1/7)	Haman Fm.	ss/sh
088	AF	36.2	128.8	10	88.2	66.8	26.3	58.0	8.2	36.1	15.6	69.1	204.2	8.8	12.0	N	[6]	106.4	1.2	(1/8)	Sagok Fm.	Red sh
089	47	36.2	128.8	13	17.0	34.6	28.0	49.4	5.7	54.7	8.8	6.9	225.2	5.0	7.6	N	[9]	106.4	1.2	(1/8)	Sagok Fm.	ss/sh
090	Hakbo	35.9	128.6	12	26.0	46.0	42.5	54.6	10.1	19.6	17.6	55.7	207.1	10.1	14.3	N	[15]	107.0	0.2	(1/1)	Hakbong Fm.	Volcanic rock
091	9	36.1	128.1	12	13.1	40.2	27.4	55.3	8.7	26.0	15.4	67.9	210.7	8.8	12.4	N	[11]	107.2	0.2	(4/5)	Shilla Cogl. Fm.	Red slts
092	K18	36.2	128.7	4	27.0	63.0	36.0	58.0	5.2	312.8	9.8	61.5	202.8	5.6	7.7	N	[7]	107.3	0.5	(2/2)	Jeomgok Fm.	ss/sh
093	Silla	35.9	128.6	15	21.3	53.8	47.3	62.7	5.7	45.7	12.5	53.4	190.8	7.0	8.9	N	[15]	107.4	0.8	(1/1)	Shilla Cogl. Fm.	Red bed
094	48	36.4	128.7	10	12.6	57.3	24.3	57.2	1.9	678.1	3.5	70.6	207.0	2.0	2.8	N	[9]	107.7	0.2	(2/5)	Jeomgok Fm.	ss/sh
095	AD	36.3	128.7	9	39.3	63.3	41.5	55.3	10.4	25.3	18.5	56.7	206.8	10.6	14.8	N+R	[6]	107.8	0.5	(1/2)	Jeomgok Fm.	bl sh
096	17	36.2	128.9	6	1.2	67.5	25.8	57.6	3.7	334.9	6.9	69.8	210.7	4.0	5.8	N	[7]	107.8	0.1	(2/7)	Jeomgok Fm.	ss/sh
097	48	36.3	128.8	14	24.2	45.6	41.0	60.3	1.9	464.0	3.8	58.0	197.2	2.2	2.9	N	[7]	107.9	0.1	(1/9)	Jeomgok Fm.	ss/sh
098	40	36.3	128.7	10	18.5	55.3	34.7	56.4	1.1	1781.3	2.0	62.3	206.9	1.1	1.6	N	[9]	107.9	0.1	(1/8)	Jeomgok Fm.	ss/sh
099	41	36.3	128.7	12	23.7	44.0	40.4	65.9	1.8	593.1	4.4	58.3	184.0	2.4	2.9	N	[9]	107.9	0.1	(1/8)	Jeomgok Fm.	ss/sh
100	12	35.8	128.7	17	2.3	53.7	22.6	58.0	1.7	443.0	3.2	71.8	202.9	1.8	2.5	N	[11]	108.1	0.1	K–Ar, wr	Hakbong Volcanic Fm.	Diabase flow
101	11	35.8	128.7	17	16.1	51.6	26.1	58.8	4.4	67.0	8.5	69.1	200.5	4.9	6.5	N	[11]	108.1	0.1	K–Ar, wr	Hakbong Volcanic Fm.	Diabase flow
102	13	35.8	128.7	14	10.4	55.6	46.8	55.7	5.6	51.5	10.0	52.5	203.8	5.7	8.0	N	[11]	108.1	0.1	K–Ar, wr	Hakbong Volcanic Fm.	Diabase flow
103	881	35.8	128.4	10	4.8	44.8	30.9	58.0	5.0	94.3	9.5	65.3	202.4	5.4	7.4	N	[13]	108.3	1.1	(3/3)	Chilgok Fm.	Red ss/sh/ms
104	882	35.8	128.4	11	6.6	53.2	37.4	63.5	5.0	85.8	11.3	60.4	188.5	6.3	7.9	N	[13]	108.3	1.1	(3/3)	Chilgok Fm.	Red ss/sh/ms
105	880	35.8	128.4	10	9.1	37.5	32.3	56.6	5.8	70.6	10.6	64.1	205.7	6.1	8.4	N	[13]	108.3	1.1	(3/3)	Chilgok Fm.	Red ss/sh/ms
106	887	35.8	128.4	11	12.6	41.8	33.7	61.1	4.6	100.6	9.6	63.2	194.3	5.4	7.1	N	[13]	108.6	1.6	(2/2)	Chilgok Fm.	Red ss/sh/ms
107	888	35.8	128.4	6	5.5	53.5	26.8	58.5	5.8	136.4	11.2	68.5	201.2	6.4	8.6	N	[13]	108.6	1.6	(2/2)	Chilgok Fm.	Red ss/sh/ms
108	AR	36.2	128.7	7	36.2	58.4	43.1	53.2	5.1	138.7	8.5	54.9	209.8	4.9	7.1	N	[6]	108.8	0.5	(1/2)	Gugyedong Fm.	Red sh
109	869	35.7	128.4	9	346.6	61.3	23.0	61.9	5.5	87.1	11.7	70.8	188.7	6.6	8.5	N	[13]	109.0	0.8	(3/4)	Chilgok Fm.	Red ss/sh/ms
110	870	35.7	128.4	6	359.2	49.5	26.1	49.1	6.6	103.3	10.1	67.4	225.6	5.8	8.7	N	[13]	109.0	0.8	(3/4)	Chilgok Fm.	Red ss/sh/ms
111	52	36.2	128.7	9	4.5	64.9	19.8	59.8	1.9	725.5	3.8	73.9	196.6	2.2	2.9	N	[7]	109.1	0.3	(2/5)	Hupyeongdong Fm.	ss/sh
112	AT	36.2	128.7	8	25.7	64.6	30.4	62.1	4.8	134.1	10.3	65.7	192.1	5.8	7.5	N	[6]	109.1	0.8	(1/2)	Hupyeongdong Fm.	Red sh
113	51	36.2	128.7	7	24.9	49.0	36.8	52.8	5.6	116.9	9.3	59.8	213.1	5.3	7.7	N	[7]	109.1	0.3	(2/6)	Hupyeongdong Fm.	ss/sh

(*continued on next page*)

Table 2 (*continued*)

Site		Site sampled			Geographic		Stereographic		Parameters			VGP position					Ref	Model or radiometric			Formation Name	Rock type
List	Name	Lat(N)	Long(E)	n_b	D_g	I_g	D_s	I_s	α_{95}	k	Δd	Lat(N)	Long(E)	dp	dm	(N/R)	[no.]	Age(Ma)	±	Method (m/n)		
114	MY14	35.5	128.2	9	357.4	50.1	28.3	65.7	5.8	78.7	14.2	65.7	178.7	7.7	9.4	N	[8]	109.4	3.2	(1/1)	Chilgok Fm.	ss/sh
115	K09	35.6	128.3	7	17.7	42.8	43.0	55.5	9.5	41.6	16.9	55.4	204.6	9.7	13.6	N	[8]	109.4	3.2	(1/1)	Chilgok Fm.	ss/sh
116	Site 3	35.2	128.2	27	–	–	27.8	58.7	1.9	210.8	3.7	67.5	198.8	2.1	2.8	N	[12]	109.4	3.2	(1/1)	Chilgok Fm.	Red sh
117	884	35.8	128.4	10	8.2	47.7	32.0	47.7	3.5	191.3	5.2	62.2	223.8	3.0	4.6	N	[13]	109.4	1.1	(2/3)	Chilgok Fm.	Red ss/sh/ms
118	868	35.7	128.4	10	359.9	58.0	24.6	53.7	4.2	131.6	7.1	69.8	215.3	4.1	5.9	N	[13]	109.4	0.6	(3/5)	Chilgok Fm.	Red ss/sh/ms
119	885	35.8	128.4	8	350.6	52.4	25.2	57.7	4.3	166.7	8.1	69.8	203.6	4.6	6.3	N	[13]	109.4	1.1	(2/3)	Chilgok Fm.	Red ss/sh/ms
120	883	35.8	128.4	9	14.0	52.7	36.2	58.9	4.3	141.7	8.3	61.3	199.7	4.8	6.4	N	[13]	109.4	1.1	(2/3)	Chilgok Fm.	Red ss/sh/ms
121	866	35.7	128.4	10	8.8	43.3	27.8	51.4	5.5	79.5	8.8	66.7	219.3	5.1	7.5	N	[13]	109.4	0.6	(3/5)	Chilgok Fm.	Red ss/sh/ms
122	867	35.7	128.4	9	355.7	41.4	24.4	53.8	5.9	77.5	10.0	70.0	215.2	5.8	8.2	N	[13]	109.4	0.6	(3/5)	Chilgok Fm.	Red ss/sh/ms
123	886	35.8	128.4	9	18.4	53.3	31.6	57.6	6.9	56.4	13.0	64.8	203.4	7.4	10.1	N	[13]	109.4	1.1	(2/3)	Chilgok Fm.	Red ss/sh/ms
124	Chilgo	35.9	128.5	(17)	6.4	49.0	28.9	61.5	2.9	152.8	6.1	66.8	192.8	3.4	4.5	N	[14]	109.4	3.2	(1/1)	Chilgok Fm.	Sedimentary rock
125	864	35.7	128.4	10	9.0	51.7	28.5	58.5	4.6	111.2	8.8	67.2	201.0	5.1	6.8	N	[13]	109.7	0.5	(3/6)	Chilgok Fm.	Red ss/sh/ms
126	865	35.7	128.4	10	43.2	58.8	33.5	58.0	4.8	102.4	9.1	63.3	201.9	5.2	7.1	N	[13]	109.7	0.5	(3/6)	Chilgok Fm.	Red ss/sh/ms
127	905	35.2	128.2	13	14.0	58.9	28.3	57.3	2.4	289.4	4.4	67.2	203.0	2.6	3.5	N	[13]	109.9	0.5	(3/7)	Chilgok Fm.	Red ss/sh/ms
128	906	35.2	128.2	8	21.4	56.3	39.3	55.6	5.0	125.3	8.9	58.3	204.8	5.1	7.1	N	[13]	109.9	0.5	(3/7)	Chilgok Fm.	Red ss/sh/ms
129	CG	35.0	128.2	13	18.6	55.4	35.0	53.5	5.5	49.9	9.3	61.3	209.9	5.3	7.6	N	[5]	110.0	0.6	(2/5)	Chilgok Fm.	ss/sh
130	MY15	35.5	128.2	8	14.3	56.7	37.9	57.8	5.8	91.7	10.9	59.8	201.1	6.3	8.5	N	[8]	110.2	1.6	(1/2)	Chilgok Fm.	ss/sh
131	07	36.0	128.5	8	13.8	43.6	44.6	58.7	7.3	59.0	14.2	54.9	199.1	8.1	10.9	N	[11]	110.2	1.6	(1/2)	Chilgok Fm.	Red ss/ms
132	907	35.2	128.2	5	9.7	57.3	35.8	55.8	4.3	322.2	7.7	61.1	205.3	4.4	6.2	N	[13]	110.4	0.4	(2/8)	Chilgok Fm.	Red ss/sh/ms
133	908	35.2	128.2	10	8.6	53.9	34.1	54.8	5.0	95.8	8.7	62.3	207.9	5.0	7.1	N	[13]	110.4	0.4	(2/8)	Chilgok Fm.	Red ss/sh/ms
134	MY16	35.2	128.2	13	0.7	53.0	33.2	63.5	4.0	105.9	9.0	63.1	187.0	5.0	6.3	N	[8]	110.5	1.1	(1/3)	Chilgok Fm.	ss/sh
135	AL	36.2	128.7	7	30.2	58.8	33.5	48.1	5.9	104.6	8.9	61.0	223.1	5.1	7.8	N	[6]	110.6	0.3	(2/6)	Iljik Fm.	Red ss/sh
136	AS	36.2	128.7	9	44.3	65.9	41.7	61.8	9.7	29.0	20.9	57.5	193.5	11.7	15.1	N	[6]	110.7	0.2	(2/8)	Iljik Fm.	Yellow ss
137	53	36.2	128.7	8	25.4	56.4	35.9	62.3	2.6	464.5	5.6	61.7	192.4	3.2	4.1	N	[7]	110.7	0.2	(2/8)	Iljik Fm.	ss/sh
138	06	35.9	128.6	11	8.3	56.2	36.5	65.9	5.4	73.0	13.3	60.6	182.5	7.2	8.8	N	[11]	110.7	0.6	(1/5)	Chilgok Fm.	Red ss/ms
139	08−	35.9	128.5	4	198.4	−52.9	225.8	−65.4	4.9	365.0	11.8	54.5	185.0	6.4	7.9	R	[11]	110.8	0.4	(1/8)	Chilgok Fm.	Red ss/ms
140	08+	35.9	128.5	9	13.1	53.5	42.3	66.9	6.5	64.0	16.8	56.6	181.1	8.9	10.8	N	[11]	110.8	0.4	(1/8)	Chilgok Fm.	Red ss/ms
141	MY30	35.3	128.2	11	23.0	64.7	31.7	58.3	3.8	147.3	7.2	64.6	200.4	4.2	5.6	N	[8]	110.8	0.4	(1/9)	Chilgok Fm.	ss/sh
142	MY29	35.3	128.2	12	23.4	61.7	33.6	57.9	7.0	39.4	13.3	63.1	201.2	7.6	10.3	N	[8]	110.8	0.4	(1/9)	Chilgok Fm.	ss/sh

143	57	36.3	128.7	11	4.8	41.2	13.6	44.8	12.0	15.4	17.0	74.8	255.4	9.6	15.1	N	[7]	110.9	0.2	(1/9)	Iljik Fm.	ss/sh
144	37	36.7	128.3	20	43.0	55.5	24.1	48.7	3.6	83.5	5.5	68.7	230.6	3.1	4.7	N	[11]	110.9	0.2	(1/8)	Iljik Fm.	Red bed
145	Site 2	35.2	128.1	31	–	–	53.1	60.1	7.5	12.9	15.2	48.4	193.2	8.6	11.4	N	[12]	113.5	5.0	(1/1)	Jinju Fm.	Gray ss
146	Jinju	35.8	128.4	(12)	21.8	67.8	44.0	64.6	6.5	55.6	15.3	55.7	186.4	8.4	10.4	N	[14]	113.5	5.0	(1/1)	Jinju Fm.	Sedimentary rock
147	MY34	35.3	128.2	7	20.7	54.7	34.0	52.4	7.8	61.0	12.9	61.9	213.0	7.4	10.7	N	[8]	114.1	1.3	(2/4)	Jinju Fm.	ss/sh
148	K10	35.7	128.3	10	17.9	61.1	24.2	67.2	10.0	24.8	26.6	67.3	171.5	13.8	16.6	N	[8]	114.5	1.0	(2/5)	Jinju Fm.	ss/sh
149	03	35.5	129.2	12	23.9	66.0	44.8	50.3	4.6	90.5	7.2	52.6	212.8	4.1	6.2	N	[11]	114.5	1.0	(2/5)	Jinju Fm.	Gray ss
150	CJ	35.4	127.2	21	193.1	−50.5	205.2	−53.5	4.3	55.2	7.2	69.3	213.6	4.2	6.0	R	[5]	115.5	1.0	(1/5)	Jinju Fm.	ss/sh
151	MY13	35.5	128.2	11	7.9	53.6	30.7	56.7	5.3	75.0	9.7	65.3	204.8	5.6	7.7	N	[8]	115.7	0.6	(1/9)	Jinju Fm.	ss/sh
152	Site 1	35.2	128.3	20	–	–	18.8	58.8	8.3	85.0	16.2	74.5	196.1	10.1	12.9	N	[12]	118.8	5.5	(1/1)	Hasandong Fm.	ss
153	HS	34.9	128.2	23	33.2	55.1	42.5	52.6	3.8	65.3	6.3	55.0	208.5	3.6	5.2	N + R	[5]	119.2	0.9	(3/6)	Hasandong Fm.	ss/sh
154	MY06	35.4	128.2	7	3.9	58.1	36.8	61.5	9.5	41.1	20.2	60.8	192.7	11.3	14.6	N	[8]	119.4	1.4	(2/4)	Hasandong Fm.	ss/sh
155	MY08	35.4	128.2	7	20.2	56.5	55.2	67.9	6.6	83.7	17.8	48.0	178.9	9.3	11.1	N	[8]	120.1	0.9	(2/6)	Hasandong Fm.	ss/sh
156	MY11	35.4	128.2	7	20.0	60.7	48.6	55.8	7.9	59.6	14.2	51.0	202.1	8.1	11.3	N	[8]	120.3	0.8	(2/7)	Hasandong Fm.	ss/sh
157	MY09	35.4	128.2	6	6.3	51.1	39.1	56.0	8.8	59.5	15.9	58.6	204.4	9.1	12.6	N	[8]	120.3	0.8	(2/7)	Hasandong Fm.	ss/sh
158	K11	35.6	128.3	8	17.5	43.8	43.5	54.5	11.5	24.2	20.1	54.8	206.2	11.4	16.2	N	[8]	120.3	0.8	(2/7)	Hasandong Fm.	ss/sh
159	Nagdo	35.9	128.3	(8)	6.5	60.8	38.6	64.6	7.0	62.6	16.5	59.4	186.0	9.0	11.2	N	[14]	123.5	4.0	(1/1)	Nakdong Fm.	Sedimentary rock
160	MY02	35.4	128.2	9	15.8	45.4	40.3	53.3	12.5	18.0	21.2	57.0	209.0	12.1	17.4	N	[8]	124.5	0.7	(2/6)	Nakdong Fm.	ss/sh
161	MY05	35.4	128.2	6	11.1	47.9	42.6	59.1	12.2	30.9	24.3	56.3	197.6	13.6	16.2	N	[8]	125.3	0.4	(1/9)	Nakdong Fm.	ss/sh
162	MY01	35.4	128.2	6	23.7	51.1	50.1	54.3	7.3	85.5	12.6	49.5	203.9	7.2	10.3	N	[8]	125.3	0.5	(1/8)	Nakdong Fm.	ss/sh
163	ND	34.7	128.3	13	19.6	54.2	*39.8*	*55.2*	5.3	62.7	9.3	57.7	204.8	5.3	7.5	N + R	[5]	125.3	0.5	(1/8)	Nakdong Fm.	ss/sh

Ref. [no.]: 1, Kang et al. (2000); 2, Kang and Kim (2000); 3, Kang et al. (1996); 4, Kim (1988); 5, Kim et al. (1993a); 6, Kim et al. (1993b); 7, Doh and Kim (1994); 8, Doh et al. (1994); 9, Suk and Doh (1996); 10, Kim and Kim (1991); 11, Lee et al. (1987); 12, Otofuji et al. (1983); 13, this study; 14, Kim and Jeong (1986); 15, Zhao et al. (1999).

[a] Shin and Jin (1995).
[b] Kim et al. (1997).
[c] Lee and Ueda (1976).
[d] Otofuji et al. (1983).
[e] Chough and Sohn (2010).
[f] Kang et al. (2000).
[g] Jwa et al. (2009).

and Jin (1995) compiled the published radiometric age data of the igneous rocks in the Korean Peninsula. The age of the Yucheon and Bulguksa granite ranges from Late Cretaceous to Paleogene (c. 87–50 Ma), indicating the latest formational stage of the Gyeongsang Supergroup. We revisited all the selected igneous paleomagnetic sites in order to fix the closest radiometric ages of the igneous units based on the precise sampling sites (Table 2).

5. Result: magnetostratigraphy of the Cretaceous Gyeongsang Basin

The objectives of this study are (1) to carry out a tectonic consistence test for Korean Peninsula in Cretaceous, (2) establish a preliminary magnetostratigraphy, and (3) reveal a detailed declination and inclination changes during the Cretaceous. A total of 163 sites acquired from the Cretaceous sequences are selected from the western Gyeongsang Basin. The age model in this study is established by combining published paleomagnetic data and recent U–Pb radiometric data on the stratigraphic sequence using the Mesozoic geomagnetic time scale (Walker and Geissman, 2009). The preliminary magnetostratigraphy of the Gyeongsang Basin is summarized in Fig. 6.

Calculations for every 20-my-long time window, the seven time-averaged Fisherian mean directions and poles from 50 Ma to 130 Ma are computed to drive the multiples of about 10-Ma interval (Fig. 7), which define the APWP (apparent polar wander path) for the Korean Peninsula during Cretaceous.

We obtained the APWP positions for the mean times as shown in Fig. 7 and listed in Table 3: (1) 212.0°E/81.3°N ($A_{95}=4.1°$), 66.3 ± 3.7 Ma; (2) 219.8°E/76.4°N (4.2°), 71.8 ± 4.2 Ma; (3) 215.2°E/70.7°N (3.9°), 79.8 ± 5.5 Ma; (4) 212.8°E/70.2°N (2.6°), 90.9 ± 5.4 Ma; (5) 207.3°E/66.7°N (2.1°), 103.8 ± 5.9 Ma; (6) 203.6°E/64.1°N (2.0°), 108.6 ± 3.7 Ma; (7) 199.9°E/60.4°N (2.9°), 115.2 ± 5.3 Ma.

6. Discussion

The paleo-magnetostratigraphic investigations reported in the present work reveal a dominantly normal polarity for the Shindong and Hayang groups and the lower part of the Yucheon Group, belonging to the Cretaceous Long Normal Superchron (KLNS, C34N, 84.0–124.6 Ma), whereas the upper part of the Yucheon Group and the Bulguksa Granites show dual polarities after the KLNS.

Notable time-declination changes are found in the APWP in Figs. 6 and 7. Although a declinational deviation of about 30° between the Lower and Upper Cretaceous rocks in the Gyeongsang Basin has been previously recognized (Kim et al., 1993b; Zhao et al., 1999; Park et al., 2005), the time-scaled magnetostratigraphy and APWP in Korea provide further insights into the tectonic evolution of East Asia during the Cretaceous. Three phases of tectonic activity during the Cretaceous are defined by comparing the concordance of paleopoles from Eurasia, China, and Korea. These might offer important clues to the post-collision process in East Asian region, as briefly discussed below.

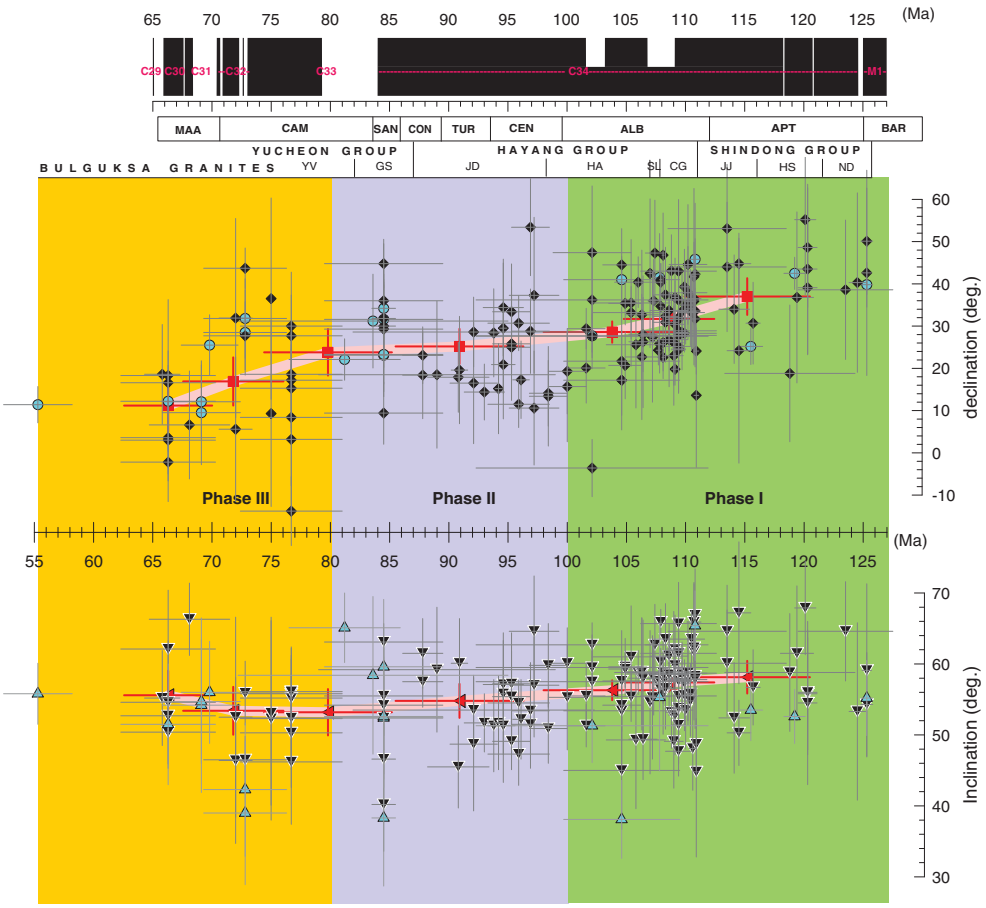


Fig. 6. Summary of preliminary magnetostratigraphy with the geomagnetic polarity timescale (Walker and Geissman, 2009) in the Gyeongsang Basin. Declination (gray circle) and inclination (gray triangle) plots are associated with vertical ($\Delta d = \sin^{-1}(\sin \alpha_{95}/\cos I)$) and α_{95}) and horizontal (age) error limits. Sky-blue symbol denote data including reversed polarity. Red symbols (square and triangle) and line display mean declination and inclination respectively. Abbreviations: ND, Nagdong Formation; HS, Hasandong Formation; JJ, Jinju Formation; CG, Chilgok Formation; SL, Shilla Conglomerate Formation; HA, Haman Formation; JD, Jindong Formation; GS, Gogeong Formation; YV, Yucheon Volcanic complex; BAR, Barremian; APT, Aptian; ALB, Albian; CEN, Cenomanian; TUR, Turonian; CON, Coniacian; SAN, Santonian; CAM, Campanian; MAA, Maastrichtian.

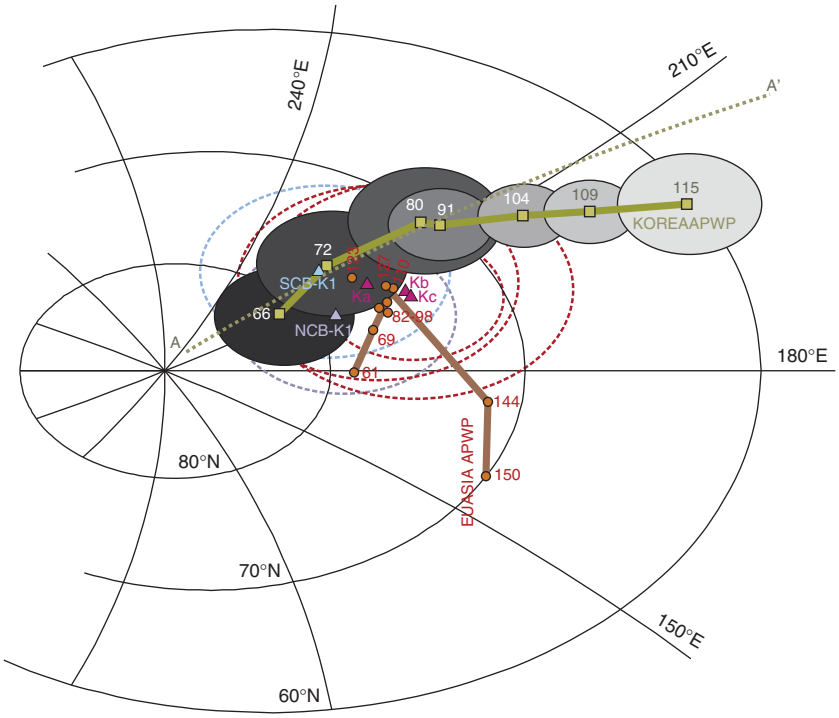


Fig. 7. Comparison of Cretaceous APWPs for Korea (golden square, this study), for Eurasia (orange circle, Besse and Courtillot, 1991) and for China (red triangle, Cretaceous mean pole of United China (Wu et al., 1993; Ma et al., 1993; Gilder and Courtillot, 1997); violet triangle, Early Cretaceous mean pole of North China Block (Zhao et al., 1999); sky blue triangle, Early Cretaceous mean pole of South China Block (Besse and Courtillot, 1991)) associated with 95% confidence level. Number next to each symbol is the mean age. Note the systematic APWP migration of Korea APWP during 120–100 Ma, which is a distinct feature as compared to the clustering of contemporary APWPs of Eurasia and China. The curve AA' (golden) is the expected paleolatitude (36°N) for the Gyeongsang Basin (35.6°N 128.6°E) in Late Cretaceous. This suggests that Korea rotated clockwise with a southward migration during Early Cretaceous (Phase I).

6.1. Phase I (100–130 Ma)

The APWP of the Korean Peninsula (Fig. 7) shows remarkable declination changes of 15.0° from 115 Ma to 91 Ma, distinguished from the APWP of Eurasia, which is characterized by a standstill between 130 and 70 Ma (Kent et al., 1986; Besse and Courtillot, 1991; Enkin et al., 1992; Gilder and Courtillot, 1997). Combining the APWP for Korea and the master Polar Wander Path for Eurasia (Besse and Courtillot, 1991), the time-declination change is interpreted to be due to the vertical axis clockwise rotation of Korea with respect to Eurasia during 130 to 100 Ma (Phase I). We notice that the amount of rotation angle in the interval from 115.2 to 103.8 Ma is 11.8° (mean rotation: 0.74°/Ma), and that the amount of paleolatitude change in the same interval is 3.5° (Fig. 7). Southward migration of the Korean Peninsula may have been associated with the APWP or inclination data in this time interval, although it is not obviously distinguished at the 95% confidence level. The Gyeongsang Basin was rifted and extended, but without spreading during the formation of the Shindong and Hayang Groups (Chough and Sohn, 2010). This timing of the basin formation is interpreted to be coeval with the Phase I proposed in this study (Fig. 6).

Gilder and Courtillot (1997) proposed that the UCB (United China Block), which is equivalent to the NCB (North China Block) plus SCB (South China Block), was completely amalgamated to the MON (Mongolia Block) in the Early Cretaceous, because Cretaceous poles of both UCB and MON are statistically identical, consistent with the hairpin-turn hypothesis on the Eurasian APWP between 130 and 70 Ma (Besse and Courtillot, 1991).

Using paleomagnetic data, Lee et al. (1992, 1997, 1999) recognized the possible extension of the collision zones between NCB and SCB to Korea and Japan, and demonstrated oblique collision (Folding-Ruler model, see Fig. 8) of the SKB (Sino-Korean Block including China, Korea, and Japan) with respect to paleo-Eurasia continent during the Mesozoic (Lin et al., 1985; Zhao et al., 1990). Comparison of stratigraphy, paleolatitude change, and fossil record indicates that NCB (North China Block) and YNB (Yeongnam Block) have been part of the NSK (North Sino-Korea Block; see Fig. 8) since the Paleozoic (Lee, 1987; Lee et al., 1999).

Table 3
Cretaceous APWP (apparent polar wander path) for Korea (data list in Table 2).

Mean age (Ma)	Number of sites	Direction		Parameters		VGP position				Mean location		Interval (Ma)	Reference no. (see Table 2)
		Dec	Inc	α_{95}	k	Lat(N)	Long(E)	A_{95}	K	Lat(N)	Long(E)		
66.3 ± 3.7	12	11.2	55.6	3.5	158.4	81.3	212.0	4.1	110.9	35.8	129.0	50–70	2, 3, 6, 12
71.8 ± 4.2	27	16.9	53.4	3.4	68.8	76.4	219.8	4.2	44.8	35.6	128.8	60–80	2, 3, 6, 10, 12
79.8 ± 5.5	31	23.8	53.2	3.3	63.6	70.7	215.2	3.9	44.5	35.4	128.7	70–90	1, 2, 3, 4, 10
90.9 ± 5.4	37	25.2	54.8	2.4	98.2	70.2	212.8	2.6	82.7	35.6	128.8	80–100	1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 14
103.8 ± 5.9	85	28.6	56.3	1.4	124.0	66.7	207.3	2.1	55.1	35.8	128.6	90–110	1, 3, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15
108.6 ± 3.7	89	31.7	57.4	1.3	128.2	64.1	203.6	2.0	56.3	35.8	128.5	100–120	3, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15
115.2 ± 5.3	35	37.0	58.1	2.3	114.2	60.4	199.9	2.9	72.9	35.6	128.3	110–130	5, 6, 7, 8, 11, 12, 13, 14

The collision of the SKB (Fitches and Zhu, 2006; Oh, 2006; Kim et al., 2009, 2011) with the continent initiated at the western part, and the SKB was continuously amalgamated to the east. The hairpin-turn hypothesis (Gilder and Courtillot, 1997) implies the complete amalgamation of the SKB to the Laurasia megacontinent in the Early Cretaceous at about 130 Ma (Fig. 7). This assumption is supported by the paleo-stress field study (Hwang, 1994) in Korea during the Early Cretaceous. We interpret that the eastern part of the SKB was located in the regional N–S compression field during the amalgamation process, and possibly underwent southward migration (Fig. 8). The sinistral NE-trending faults developed in the SKB such as the Tan-Lu, the Chugaryeong, and the Okcheon-Boundary Fault Systems (Fig. 9). The declination of the Early Cretaceous to Late Jurassic rocks in the eastern part of Tan-Lu fault system possess a clockwise deviation with respect to the western part (Fig. 9). Zhu (1993) analyzed paleomagnetic data and bending geometry along the Tan-Lu Fault, and suggested that the northern section of the fault has rotated clockwise about 20° relative to the southern section (Fig. 9). An alternate interpretation may be that the rigid block rotation of the entire eastern part of the Tan-Lu fault has taken place along the fault.

We propose this previously unrecognized disturbance of Phase I as "Goguryeo Disturbance", after the name of an ancient kingdom that subjugated most of Korea and Manchuria during 37 BC and AD 668. Faulting, rifting, basin development, and igneous activity in the intra-continental area in the Sino-Korea Block may have been responsible for the Goguryeo Disturbance.

6.2. Phase II (80–100 Ma)

During Phase II (80–100 Ma), the arc volcanic activity represented by the Yucheon Group reached the peak in the backarc basin (Chough and Sohn, 2010). The APWP positions for Korea are statistically clustered in this time interval similar to the contemporary poles for Eurasia or China, and no declination change is observed. Gilder and Courtillot (1997) produced an averaged pole of the Cretaceous for the united China block (UCB = NCB + SCB) at 201.0°E/74.5°N (A_{95} = 4.7°). However, comparison of the poles between Korea (212.8°E/70.2°N (A_{95} = 2.6°) for 90.9 ± 5.4 Ma and 215.2°E/70.7°N (A_{95} = 3.9°) for 79.8 ± 5.5 Ma), China and Eurasia (197.1°E/76.7°N (A_{95} = 5.0°) for 100 Ma, 200.1°E/76.7°N (A_{95} = 3.2°) for 90 Ma and 198.9°E/76.2°N (A_{95} = 3.4°) for 80 Ma) indicates possible

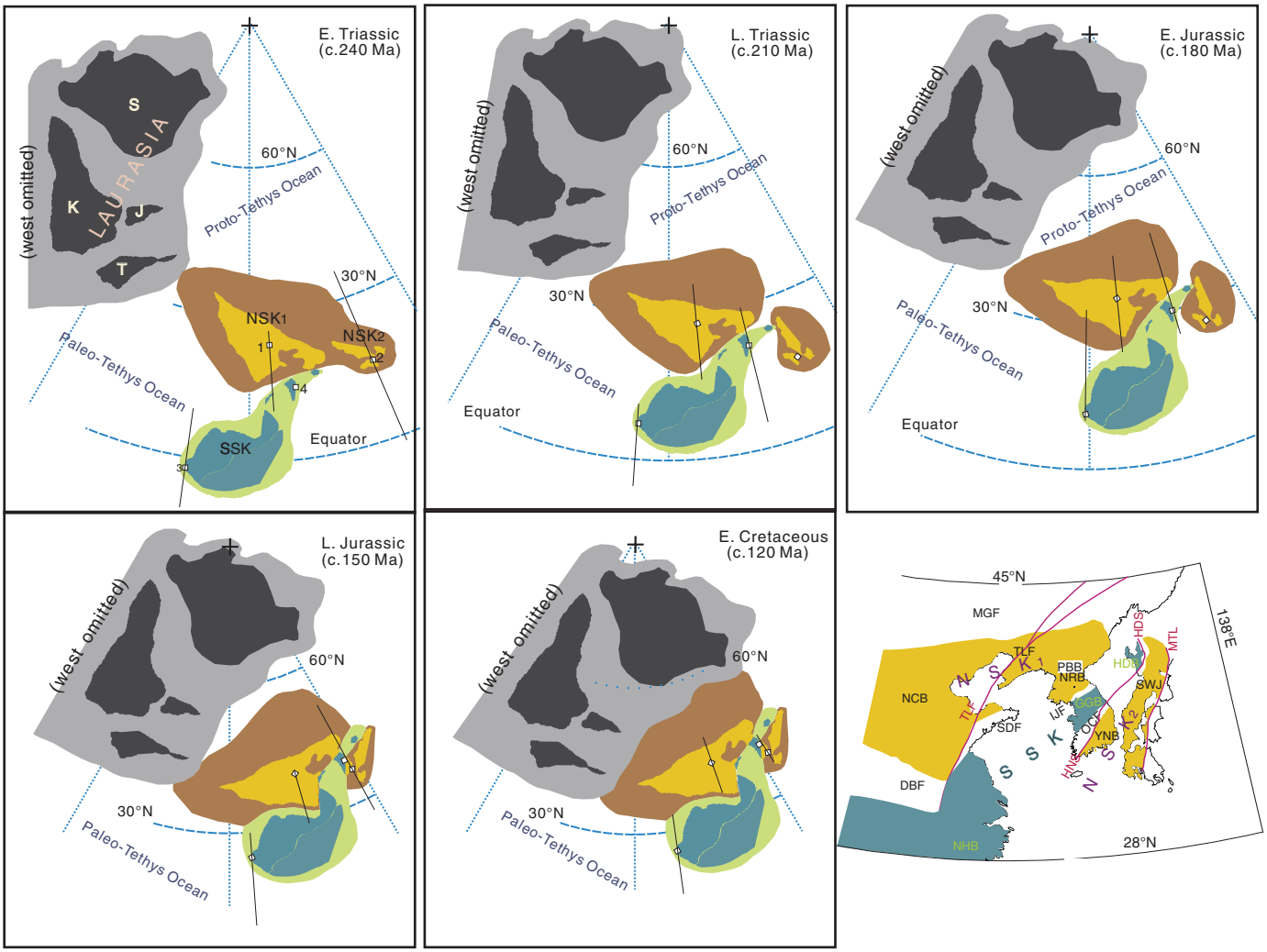


Fig. 8. Time–space first-order reconstruction map in East Asia region from the collision stage (A) in Early Triassic to the amalgamation stage (E) in Early Cretaceous, which is modified after folding-ruler-model (Lee et al., 1992, 1997, 1999). A bar in open square is uncertainty in paleo-latitude for each reference point calculated from paleomagnetic data. The Late Mesozoic tectonic index map (F) represent three micro-blocks of NSK_1 (North Sino-Korea Block 1), NSK_2 (North Sino-Korea Block 2) and SSK (South Sino-Korea Block). NSK_1 = NCB (North China Block) + NRB (Nangrim Block, North Korea); NSK_2 = YNB (Yeongnam Block) + SWJ (South-west Japan Block); SSK = SCB (South China Block = Yangtzu + Nanhua blocks) + GGB (Gyeonggi Block) + HDB (Hida Block). Open squares (1–4) in A to E denote paleomagnetic reference points: 1. Beijing (40°N, 116°E) for NSK_1, 2. Busan (35°N, 129°E) for NSK_2, 3. Kunming (25°N, 103°E) for SSK (China), 4. Chungju (37°N, 128°E) for SSK (Korea). DBF, Dabie Foldbelt; IJF, Imjingang Foldbelt; MGF, Monggolian Foldbelt; OCF, Okcheon Foldbelt; QLF, Qinling Foldbelt, SDF, Shandong Foldbelt; TLF, Tan-Lu Fault; MTL, Median Tectonic Line; HDS, Hida Shearzone; HNS, Honam Shear Zone.

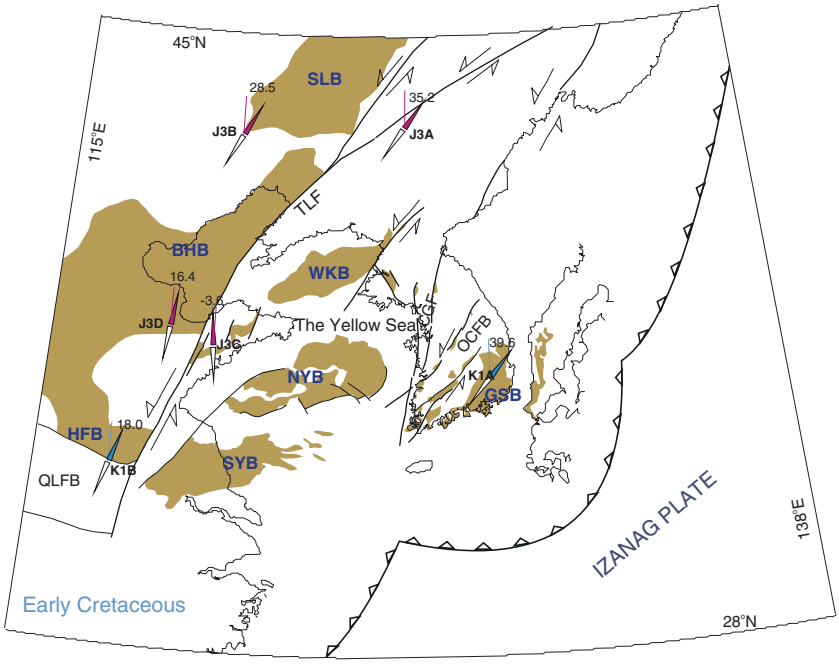


Fig. 9. Late Mesozoic Paleomagnetic directions for main part of the Sino-Korea Block plotted on the Early Cretaceous reconstruction map (Lee et al., 1999). The gold-colored area represent Early Cretaceous terrestrial basin. The eastern part of the Tan-Lu Fault rotates clockwisely (migrates a few hundreds of kilometers). Note that sinistral movements of the faults occurred with the volcanic activities, which can drive counter clockwise block rotation locally intervening these faults such as micro-blocks along the Okcheon foldbelt (OCF) boundary faults. The arrow denotes probable EW stress field by collisions of microblocks and tectonic teranes. CGF, Chugaryeong Fault; GSB, Gyeongsang Basin; MTB, Mino-Tamba Belt; OCF, Okcheon Foldbelt; QLF, Qinling Foldbelt, RKB, Ryoke Belt, SAB, Sikhote-Alin Belt; TLF, Tan-Lu Fault; YSB, Yellow Sea Basin.

regional deformation of the East Asia by the collision between India and Eurasia since 50 Ma (Otofuji et al., 1990).

In phase II, A-type or I-type granites of intra-plate affinity might have formed during the syn- to post-orogenic stage in the back-arc region (Lee et al., 2010a).

6.3. Phase III (50–80 Ma)

The two populations of normal and reversed polarities are statistically antipodal during Phase III (50–80 Ma). The amount of clockwise rotation is 12.6° from 79.8 Ma to 66.3 Ma. A gradual declination change is evident during phase III, implying a vertical axis rotation for Korea.

This period is characterized by intrusion of the Bulguksa granite (I-type) in the back-arc system of the Japan-Gyeongsang arc system (Kutsukake, 2001; Ishihara, 2007), and agrees well with the timing of the Bulguksa Orogeny (Lee, 1987). We correlate the Bulguksa Orogeny with the subduction of the Pacific-Kula plate approximately 85 Ma beneath Korea at the Eurasia margin. This inference is supported by oceanic magnetic stripe patterns of the triple junction in the Chinook trough (Mark and Geoffrey, 1982).

7. Conclusion

Our study leads to the following major conclusions:

1. The magnetizations from the Gyeongsang Basin are primary, from which the Cretaceous paleopoles of the Korean Peninsula could be obtained.
2. By combining paleomagnetic and radiometric data, we provide a reliable magnetostratigraphy for the Gyeongsang Supergroup throughout the Cretaceous.
3. Remarkable time-declination changes are found in this study during Early Cretaceous, which can be interpreted as the result of the clockwise rotation of the Korean Peninsula with respect to China and Eurasia.
4. Cretaceous APWP of Korea reveals that the Korean Peninsula (eastern part of the Sino-Korean Block) might have migrated gradually about 300 km southward between 130 and 100 Ma, which is supported by the Early Cretaceous N–S compressional stress field in the Korean Peninsula.
5. These interpretations offer important clues to the post-collision process after amalgamation of the Sino-Korean Block into the Laurasia megacontinent. We suggest that the large intra-continental strike-slip faults (e.g., Tan-Lu Fault, Okcheon-Boundary Fault) may have been rejuvenated by the N–S compression stress field.

Supplementary material related to this article can be found online at doi:10.1016/j.gr.2011.05.002.

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